

Victoria Park Academy Shenzhen

CIE IGCSE Computer Science (0478) — YEAR 10 SEMESTER 1

ASSESSMENT MATERIALS & REPORTING PACK

CONTENTS

1. Quiz 1 — Databases & SQL (Week 3) 25 marks, 20 minutes
2. Quiz 2 — Boolean Logic (Week 5) 20 marks, 15 minutes
3. Quiz 3 — Sorting & Validation (Week 7) 20 marks, 15 minutes
4. Quiz 4 — Network Hardware (Week 8) 20 marks, 15 minutes
5. End-of-Semester Mock Exam — Paper 1 (Week 17) 75 marks, 1h 45m
6. Student Report Template
7. Bilingual Glossary Poster (Organised by Unit)

All answer keys and mark schemes are included after each assessment.

Teacher-only materials are marked with [TEACHER ONLY].

QUIZ 1 — DATABASES & SQL

Week 3 | 20 minutes | 25 marks

STUDENT NAME: _____ CLASS: _____ DATE: _____

INSTRUCTIONS:

- Answer ALL questions in the spaces provided.
- Calculators are NOT permitted.
- Use the SQL reference card if needed.

SECTION A: Multiple Choice (5 marks)

Choose the ONE correct answer for each question.

8. Which SQL keyword is used to retrieve data from a database?

- A) GET B) SELECT C) RETRIEVE D) FIND [1]

Answer: _____

9. In a relational database, what is the purpose of a primary key?

- A) To link two tables together
B) To uniquely identify each record in a table
C) To store foreign data
D) To encrypt sensitive information [1]

Answer: _____

10. Which type of relationship exists when one record in Table A can be linked to many records in Table B?

- A) One-to-one B) One-to-many C) Many-to-many [1]

Answer: _____

11. A database containing all customer information in a single table with repeated order details is an example of:

- A) A relational database B) A flat-file database
C) A normalised database D) A distributed database [1]

Answer: _____

5. Which SQL wildcard represents exactly one character?

- A) % B) * C) _ D) ? [1]

Answer: _____

SECTION B: Short Answer (10 marks)

12. State the name of the process used to organise data in a database to reduce data redundancy and improve data integrity. [1]

13. A school database stores student information. Table 1 shows a poorly designed table:

StudentID	FirstName	LastName	ClassName	TeacherName
S001	Wei	Li	10A	Mr Zhang
S002	Mei	Wang	10A	Mr Zhang
S003	Hao	Chen	10B	Mrs Liu
S004	Yun	Zhang	10A	Mr Zhang
S005	Kai	Huang	10B	Mrs Liu

(a) Identify TWO examples of data redundancy in this table. [2]

Example 1: _____

Example 2: _____

(b) Explain how splitting this table into two separate tables would improve the database design. [2]

14. Write a SQL query to display the FirstName and LastName of all students in Class 10A from a table called Students. [2]

15. Explain the difference between a primary key and a foreign key. [3]

Primary key: _____

Foreign key: _____

SECTION C: Problem Solving (10 marks)

16. A library wants to create a database to track books and borrowers.

Each book has a BookID, Title, Author, and Genre.

Each borrower has a BorrowerID, FirstName, LastName, and Email.

A borrower can borrow many books, but each book can only be borrowed by one borrower at a time.

(a) Draw an Entity-Relationship Diagram (ERD) for this database.

Show the two entities, their attributes (underline the primary key), and the cardinality of the relationship between them. [4]

[Use the space below to draw your ERD]

(b) A librarian needs to find all books written by the author

"Liu Cixin". Write the SQL query. [2]

(c) Write a SQL query to add a new book to the database:

BookID: B105, Title: "The Three-Body Problem", Author: "Liu Cixin",
Genre: "Science Fiction" [2]

(d) The library wants to create a report showing all borrowers and the

titles of books they have borrowed. Write a SQL query using a JOIN to produce this report. [2]

END OF QUIZ 1 — Total: 25 marks

[TEACHER ONLY] QUIZ 1 — ANSWER KEY & MARK SCHEME

SECTION A: Multiple Choice (5 marks)

17.B (SELECT) — 1 mark

18.B (To uniquely identify each record in a table) — 1 mark

19.B (One-to-many) — 1 mark

20.B (A flat-file database) — 1 mark

21.C () — 1 mark

SECTION B: Short Answer (10 marks)

6. Normalisation — 1 mark

(Accept: "Normalisation" or "Database normalisation")

22. (a) Award 1 mark each for any TWO of:

- "Mr Zhang" appears three times / repeated teacher name for 10A
- "Mrs Liu" appears twice / repeated teacher name for 10B
- "10A" appears three times / class name is repeated
- Teacher name is repeated for every student in the same class

[2 marks total]

(b) Award up to 2 marks:

- 1 mark: Create a separate Class table with ClassName as primary key and TeacherName as a field
- 1 mark: The Student table then only stores ClassName (foreign key), eliminating repeated TeacherName data / reducing redundancy
- MAX 1 mark if only says "split into two tables" without explanation

[2 marks total]

23. Award 1 mark for SELECT with correct columns; 1 mark for FROM and WHERE:

SELECT FirstName, LastName -- 1 mark

FROM Students -- 0.5 mark

WHERE ClassName = '10A'; -- 0.5 mark

(Accept: WHERE Class = '10A' or equivalent column name)

[2 marks total]

9. Award up to 3 marks:

- 1 mark: Primary key is a field (or combination of fields) that uniquely identifies each record in a table
- 1 mark: Foreign key is a field in one table that refers to the primary key in another table
- 1 mark: Used to create a link/relationship between two tables

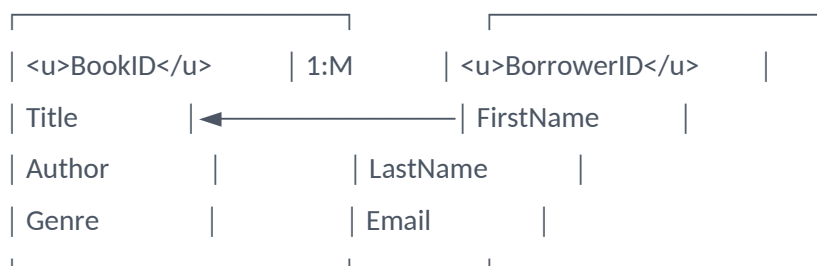
[3 marks total]

SECTION C: Problem Solving (10 marks)

24.(a) ERD — 4 marks total:

Entity: BOOK

Entity: BORROWER



Mark allocation:

- 1 mark: Both entities (BOOK and BORROWER) correctly named
- 1 mark: Correct attributes listed with BookID and BorrowerID underlined as primary keys
- 1 mark: Relationship line drawn between entities with correct direction
- 1 mark: Correct cardinality (1:M) shown — one borrower can borrow many books

[4 marks total]

(b) Award 1 mark for correct SELECT/FROM; 1 mark for correct WHERE:

SELECT * FROM Books -- 1 mark

WHERE Author = 'Liu Cixin'; -- 1 mark

(Accept: SELECT Title FROM Books WHERE Author = 'Liu Cixin')

[2 marks total]

(c) Award 1 mark for correct INSERT INTO syntax; 1 mark for VALUES:

INSERT INTO Books (BookID, Title, Author, Genre)

VALUES ('B105', 'The Three-Body Problem', -- 1 mark

'Liu Cixin', 'Science Fiction'); -- 1 mark

[2 marks total]

(d) Award 1 mark for JOIN syntax; 1 mark for correct ON condition:

SELECT Borrowers.FirstName, Borrowers.LastName, Books.Title

FROM Borrowers

INNER JOIN Books

ON Borrowers.BorrowerID = Books.BorrowerID; -- 2 marks

(Accept: any correct JOIN syntax with appropriate ON clause)

[2 marks total]

--- TOTAL: 25 marks ---

Grade boundaries (suggested):

A*: 23-25 A: 20-22 B: 17-19 C: 14-16 D: 11-13 E: 8-10

QUIZ 2 — BOOLEAN LOGIC

Week 5 | 15 minutes | 20 marks

STUDENT NAME: _____ CLASS: _____ DATE: _____

INSTRUCTIONS:

- Answer ALL questions in the spaces provided.
- Draw gate symbols clearly.
- Use 0 and 1 (not TRUE and FALSE) for truth table answers.

SECTION A: Multiple Choice (5 marks)

25. Which logic gate produces an output of 1 only when BOTH inputs are 1?

A) OR B) AND C) NOT D) XOR [1]

Answer: _____

2. A NOT gate has how many inputs?

A) 0 B) 1 C) 2 D) 3 [1]

Answer: _____

3. The output of a NAND gate is 0 only when:

- A) Both inputs are 0
- B) Both inputs are 1

C) The inputs are different

D) At least one input is 0 [1]

Answer: _____

26. Which gate produces an output of 1 when its two inputs are DIFFERENT?

A) AND B) OR C) XOR D) NOR [1]

Answer: _____

27. The circle symbol (o) on the output of a logic gate means:

A) The output is amplified

B) The output is inverted (NOT)

C) The gate has three inputs

D) The gate is broken [1]

Answer: _____

SECTION B: Short Answer (7 marks)

28. Complete the truth table for a NOR gate: [2]

Input A Input B Output X		
0	1	
1	0	

7. Complete the truth table for this logic expression:

$X = (A \text{ AND } B) \text{ OR } (\text{NOT } C)$ [3]

Input A Input B Input C A AND B NOT C Output X					
0	0	1			
1	1	0			
1	0	1			

8. Write the logic expression for this circuit description:

"The output is 1 when input A is 1 AND input B is NOT 1" [2]

SECTION C: Problem Solving (8 marks)

29. (a) Draw a logic circuit for the expression: $X = (A \text{ OR } B) \text{ AND } C$ [3]

Show all inputs, gates, and the output clearly.

(b) Complete the truth table for the circuit in part (a): [3]

Input A Input B Input C A OR B Output X				
0	0	1		

	0		1		1						
	1		1		0						

(c) A security light system uses two inputs:

- M (motion sensor): 1 = motion detected, 0 = no motion
- D (daylight sensor): 1 = dark, 0 = light

The light should turn ON (output = 1) only when motion is detected AND it is dark.

(i) Write the logic expression for this system. [1]

(ii) Name the single logic gate that could implement this system. [1]

END OF QUIZ 2 — Total: 20 marks

[TEACHER ONLY] QUIZ 2 — ANSWER KEY & MARK SCHEME

SECTION A: Multiple Choice (5 marks)

30.B (AND) — 1 mark

31.B (1) — 1 mark

32.B (Both inputs are 1) — 1 mark

33.C (XOR) — 1 mark

34.B (The output is inverted / NOT) — 1 mark

SECTION B: Short Answer (7 marks)

6. NOR gate truth table — 2 marks (0.5 each row):

Input A	Input B	Output X
0	0	1 ← 0.5 mark
0	1	0 ← 0.5 mark
1	0	0 ← 0.5 mark
1	1	0 ← 0.5 mark

[2 marks total]

7. Truth table for $X = (A \text{ AND } B) \text{ OR } (\text{NOT } C)$ — 3 marks:

Input A	Input B	Input C	A AND B	NOT C	Output X
0	0	0	0	1	1
0	0	1	0	0	0
1	1	0	1	1	1
1	0	1	0	0	0

Mark allocation:

- 1 mark: A AND B column all correct

- 1 mark: NOT C column all correct
- 1 mark: Output X column all correct

[3 marks total]

8. $X = A \text{ AND } (\text{NOT } B)$ — 2 marks

Award 1 mark for $A \text{ AND } \dots$; 1 mark for NOT B

(Accept: $X = A \text{ AND } \bar{B}$ or $X = A \cdot \bar{B}$ or equivalent notation)

[2 marks total]

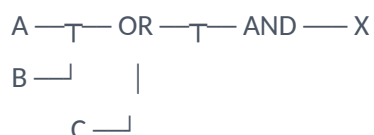
SECTION C: Problem Solving (8 marks)

35. (a) Logic circuit for $X = (A \text{ OR } B) \text{ AND } C$ — 3 marks:

Mark allocation:

- 1 mark: OR gate drawn correctly with A and B as inputs
- 1 mark: AND gate drawn correctly
- 1 mark: C connected as second input to AND gate, output labelled X correctly

Diagram description for marking:



[3 marks total]

(b) Truth table — 3 marks (1 mark per correct output column):

Input A	Input B	Input C	A OR B	Output X
0	0	0	0	← row 1: 1 mark
0	0	1	0	
0	1	1	1	← rows 2-3: 1 mark
1	1	0	1	← row 4: 1 mark

[3 marks total]

(c) (i) $\text{Light} = M \text{ AND } D$ (or $X = M \text{ AND } D$) — 1 mark

(Accept: $X = M \cdot D$)

(ii) AND gate — 1 mark

(Must match expression in (i); if they wrote M OR D, accept OR gate but no mark for (i))

[2 marks total]

--- TOTAL: 20 marks ---

Grade boundaries (suggested):

A*: 18–20 A: 16–17 B: 14–15 C: 12–13 D: 10–11 E: 8–9

Week 7 | 15 minutes | 20 marks

STUDENT NAME: _____ CLASS: _____ DATE: _____

INSTRUCTIONS:

- Answer ALL questions in the spaces provided.
- Section B requires you to complete a trace table — work carefully.
- Section C requires Python code — syntax matters.

SECTION A: Multiple Choice (5 marks)

36. In a bubble sort, how many items are compared on each pass through the list?

- A) All items are compared to every other item
- B) Each pair of adjacent items is compared
- C) Only the first and last items are compared
- D) Items are compared to a pivot value [1]

Answer: _____

37. Which validation check would ensure a password is at least 8 characters long?

- A) Range check B) Type check C) Length check D) Format check [1]

Answer: _____

38. After one complete pass of bubble sort on [5, 2, 8, 1], what is the list?

- A) [1, 2, 5, 8] B) [2, 5, 1, 8] C) [2, 1, 5, 8] D) [5, 2, 1, 8] [1]

Answer: _____

39. Which of the following is NOT a type of validation check?

- A) Range check B) Parity check C) Presence check D) Type check [1]

Answer: _____

5. Bubble sort has a worst-case time complexity of:

- A) $O(n)$ B) $O(n \log n)$ C) $O(n^2)$ D) $O(1)$ [1]

Answer: _____

SECTION B: Trace Table (7 marks)

40. Trace the following bubble sort algorithm on the array [7, 3, 9, 2].

Show the state of the array after each comparison and swap.

Pseudocode:

FOR $i \leftarrow 1$ TO 3

FOR $j \leftarrow 1$ TO $3 - i$

IF $\text{Data}[j] > \text{Data}[j + 1]$ THEN

Temp $\leftarrow \text{Data}[j]$

```

    Data[j] ← Data[j + 1]
    Data[j + 1] ← Temp
ENDIF
NEXT j
NEXT i

```

Trace table:

Pass	Data[1]	Data[2]	Data[3]	Data[4]	Swap?	
Start	7	3	9	2	—	
After Pass 1:	[_____, _____, _____, _____]					[4]
After Pass 2:	[_____, _____, _____, _____]					[2]
After Pass 3:	[_____, _____, _____, _____]					[1]

SECTION C: Code Writing (8 marks)

41. A cinema booking system needs to validate user input for:

- Number of tickets (must be between 1 and 10 inclusive)
- Email address (must contain an @ symbol)

Write a Python program that:

- Asks the user to input the number of tickets they want. [1]
- Uses a range check to validate that the number is between 1 and 10. If not, display an error message and ask again. [3]
- Asks the user to input their email address. [1]
- Uses a format check to validate that the email contains '@'. If not, display an error message and ask again. [3]

Write your answer below:

END OF QUIZ 3 — Total: 20 marks

[TEACHER ONLY] QUIZ 3 — ANSWER KEY & MARK SCHEME

SECTION A: Multiple Choice (5 marks)

- B (Each pair of adjacent items is compared) — 1 mark
- C (Length check) — 1 mark
- B ([2, 5, 1, 8]) — Working: compare 5&2→swap [2,5,8,1]; compare 5&8→no swap; compare 8&1→swap [2,5,1,8] — 1 mark
- B (Parity check) — Parity check is an error detection method, not a validation check — 1 mark
- C ($O(n^2)$) — 1 mark

SECTION B: Trace Table (7 marks)

6. Completed trace table:

Pass	Data[1]	Data[2]	Data[3]	Data[4]	Swap?	
Start	7	3	9	2	—	
1	3	7	9	2	Yes	← 7>3 swap
1	3	7	9	2	No	← 7<9 no
1	3	7	2	9	Yes	← 9>2 swap
2	3	7	2	9	No	← 3<7 no
2	3	2	7	9	Yes	← 7>2 swap
3	2	3	7	9	No	← 3>2 swap

Mark allocation:

- Pass 1: 1 mark for all three rows correct [7,3,9,2→3,7,9,2→3,7,9,2→3,7,2,9]
- Pass 2: 1 mark for both rows correct [3,7,2,9→3,7,2,9→3,2,7,9]
- Pass 3: 1 mark for row correct [2,3,7,9]
- After Pass 1: [3, 7, 2, 9] — 1 mark
- After Pass 2: [3, 2, 7, 9] — 2 marks
- After Pass 3: [2, 3, 7, 9] — 1 mark

NOTE: Some students may show intermediate swap states — accept if the final state after each pass is correct.

[7 marks total]

SECTION C: Code Writing (8 marks)

7. Mark allocation:

(a) 1 mark: Asks for number of tickets (input statement)

(b) 3 marks:

- 1 mark: Uses a range check (if statement with and/or)
- 1 mark: Correct condition (≥ 1 and ≤ 10 or equivalent)
- 1 mark: Displays error and loops/repeats if invalid

(c) 1 mark: Asks for email address (input statement)

(d) 3 marks:

- 1 mark: Uses a format check (if 'email' condition)
- 1 mark: Correct condition (checks for '@' in string)
- 1 mark: Displays error and loops/repeats if invalid

Example solution:

Part (a) — 1 mark

```
tickets = int(input("How many tickets? "))
```

Part (b) — 3 marks

```

while tickets < 1 or tickets > 10:          # 1 mark range check
    print("Error: Must be between 1 and 10")    # 0.5 error msg
    tickets = int(input("How many tickets? "))  # 0.5 repeat
# Part (c) — 1 mark
email = input("Enter your email: ")
# Part (d) — 3 marks
while "@" not in email:                    # 1 mark format check
    print("Error: Email must contain @")        # 0.5 error msg
    email = input("Enter your email: ")        # 0.5 repeat
print("Booking confirmed!")

```

Acceptable alternatives:

- Using if/else instead of while (0.5 marks max for loop logic)
- Using try/except for type checking (award marks if functional)
- Using separate if statements instead of combined condition
(full marks if logically correct)

Deduct 1 mark maximum for minor syntax errors that do not affect logic.

[8 marks total]

--- TOTAL: 20 marks ---

Grade boundaries (suggested):

A*: 18–20 A: 16–17 B: 14–15 C: 12–13 D: 10–11 E: 8–9

QUIZ 4 — NETWORK HARDWARE

Week 8 | 15 minutes | 20 marks

STUDENT NAME: _____ CLASS: _____ DATE: _____

INSTRUCTIONS:

- Answer ALL questions in the spaces provided.
- Diagrams should be drawn clearly with labels.

SECTION A: Multiple Choice (5 marks)

47. Which device connects a computer to a network using a unique hardware address burned into it at manufacture?

A) Router B) Switch C) NIC D) Access point [1]

Answer: _____

2. A MAC address consists of how many hexadecimal digits?

A) 8 B) 12 C) 16 D) 32 [1]

Answer: _____

48. In which topology does every device connect to a single central device?

- A) Bus B) Star C) Mesh D) Ring [1]

Answer: _____

4. Which of the following is a valid IPv4 address?

- A) 192.168.1.256 B) 10.0.0.1 C) 256.0.0.1 D) 172.300.1.1 [1]

Answer: _____

5. In a peer-to-peer network:

- A) All devices are equal and can act as both client and server
B) There is one powerful central server that controls everything
C) All devices connect through a single backbone cable
D) Every device is directly connected to every other device [1]

Answer: _____

SECTION B: Short Answer (7 marks)

49. (a) State the difference between a MAC address and an IP address. [2]

MAC address: _____

IP address: _____

(b) Give ONE reason why a device needs both a MAC address and an IP address to communicate on a network. [1]

50. Explain ONE advantage and ONE disadvantage of using a star topology compared to a bus topology. [4]

Advantage: _____

Disadvantage: _____

SECTION C: Design (8 marks)

51. Victoria Park Academy Shenzhen wants to set up a network for a new computer lab with 30 student computers, 1 teacher computer, and 1 network printer. The lab must share internet access and allow the teacher to monitor all student screens.

(a) Draw a network diagram showing the most appropriate topology for this lab. Label all hardware devices. [4]

(b) State TWO pieces of hardware (other than computers) that would be needed, and explain the purpose of each. [4]

Hardware 1: _____

Purpose: _____

Hardware 2: _____

Purpose: _____

END OF QUIZ 4 — Total: 20 marks

[TEACHER ONLY] QUIZ 4 — ANSWER KEY & MARK SCHEME

SECTION A: Multiple Choice (5 marks)

52.C (NIC — Network Interface Card) — 1 mark

53.B (12 hexadecimal digits, often written as 6 pairs) — 1 mark

54.B (Star topology) — 1 mark

55.B (10.0.0.1) — 1 mark

(A and D contain values > 255; C starts with 256 which is invalid)

56.A (All devices are equal and can act as both client and server) — 1 mark

SECTION B: Short Answer (7 marks)

57.(a) Award up to 2 marks:

- 1 mark: MAC address is a permanent hardware address burned into the NIC at manufacture / used for identification on a local network / 48-bit / 12 hex digits
- 1 mark: IP address is a logical address assigned by the network / used for routing across networks / can change / 32-bit (IPv4) written in dotted decimal

(Accept any correct distinguishing feature)

[2 marks total]

(b) Award 1 mark for any valid reason:

- MAC address is used for communication within the local network (data link layer)
- IP address is used for routing across different networks (network layer)
- They operate at different OSI layers
- IP addresses can change; MAC addresses are permanent

[1 mark total]

7. Advantage (2 marks):

Award 1 mark for identification; 1 mark for explanation:

- 1 mark: If one device/cable fails, the rest of the network continues to work
- 1 mark: Because each device has its own connection to the central hub/switch, so failure is isolated

OR

- 1 mark: Easy to add new devices
- 1 mark: Just connect a new cable from the new device to the

central switch

OR

- 1 mark: Centralised management / easy to monitor traffic
- 1 mark: All data passes through the central device

[2 marks total]

Disadvantage (2 marks):

Award 1 mark for identification; 1 mark for explanation:

- 1 mark: If the central hub/switch fails, the entire network fails
- 1 mark: All devices depend on this single point

OR

- 1 mark: More expensive than bus topology
- 1 mark: Requires more cable and a central device (switch/hub)

OR

- 1 mark: More cabling required than bus
- 1 mark: Each device needs its own cable to the centre

[2 marks total]

SECTION C: Design (8 marks)

58. (a) Network diagram — 4 marks:

Mark allocation:

- 1 mark: Star topology clearly shown (all devices connect to centre)
- 1 mark: Central device labelled (switch or hub)
- 1 mark: All 30 student PCs, 1 teacher PC, and 1 printer shown and labelled
- 1 mark: Router shown connecting the network to the Internet / external network

[Diagram description for verbal feedback to students:]

All 32 computers and the printer connect via individual cables to a central switch. The switch connects to a router which provides internet access. The teacher computer is shown at the centre for monitoring purposes.

[4 marks total]

(b) Hardware and purposes — 4 marks (2 marks each):

Hardware 1 (1 mark) + Purpose (1 mark):

- Switch: to connect all devices within the lab and manage data traffic between them

OR

- Router: to connect the lab network to the Internet and route data between networks

OR

- Server: to store shared resources, manage user accounts, and enable teacher monitoring

Hardware 2 (1 mark) + Purpose (1 mark):

- Any second valid device with appropriate purpose
- Must be different from Hardware 1

Acceptable answers:

- Network Interface Card (NIC): allows each computer to connect to the network
- Access point: provides wireless connectivity for mobile devices
- Firewall: protects the network from unauthorised access
- Cabling (Ethernet CAT5/CAT6): physical medium for data transmission

[4 marks total]

--- TOTAL: 20 marks ---

Grade boundaries (suggested):

A*: 18–20 A: 16–17 B: 14–15 C: 12–13 D: 10–11 E: 8–9

END-OF-SEMESTER MOCK EXAM — PAPER 1

Week 17 | 1 hour 45 minutes | 75 marks

STUDENT NAME: _____ CLASS: _____ DATE: _____

INSTRUCTIONS:

- Answer ALL questions in the spaces provided.
- This paper consists of THREE sections: A (30 marks), B (30 marks), and C (15 marks).
- Calculators are NOT permitted.
- Show all your working.

SECTION A: Short Answer Questions (30 marks)

Answer ALL questions. Write your answers in the spaces provided.

Topic: Data Representation

59. Convert the 8-bit binary number 10011011 to hexadecimal. [2]

60. An image has a resolution of 800 x 600 pixels and uses 24-bit colour. Calculate the file size of the uncompressed image in megabytes (MB). Show your working. [3]

61. Explain why lossy compression is NOT suitable for compressing a text document, but lossless compression is. [2]

Topic: Communication & Internet

62. Describe the role of DNS (Domain Name System) when a user types a URL into their web browser. [3]

63. State the difference between the Internet and the World Wide Web. [2]

64. Explain how packet switching works when sending data across the Internet. Include why this method is more reliable than circuit switching. [4]

Topic: Hardware & Software

65. (a) State the three components of the CPU and briefly describe the function of each. [3]

Component 1: _____

Function: _____

Component 2: _____

Function: _____

Component 3: _____

Function: _____

(b) Explain the purpose of cache memory in the CPU. [2]

66. Describe the difference between system software and application software. Give one example of each. [3]

Topic: Security & Ethics

67. Explain how a firewall protects a computer network. [2]

68. Describe what is meant by "encryption" and explain why it is important when sending data over the Internet. [3]

Topic: Databases

11. A relational database has two tables:

STUDENT(StudentID, FirstName, LastName, ClassID)

CLASS(ClassID, ClassName, Teacher)

(a) Identify the primary key in each table. [2]

STUDENT: _____

CLASS: _____

(b) Explain how the two tables are linked together. [2]

Topic: Boolean Logic

12. Complete the truth table for the logic expression:

$X = (A \text{ AND NOT } B) \text{ OR } (B \text{ AND } C)$ [4]

A	B	C	NOT B	A AND NOT B	B AND C	X

	0		0		1									
	0		1		0									
	0		1		1									
	1		0		0									
	1		0		1									
	1		1		0									

[Each column: 1 mark for all 8 rows correct]

SECTION A TOTAL: 30 marks

SECTION B: Structured Questions (30 marks)

Answer ALL questions.

QUESTION 13 — Databases & SQL (12 marks)

A hospital uses a relational database to manage patient records.

The database has three tables:

PATIENT(PatientID, FirstName, LastName, DateOfBirth, WardID)

WARD(WardID, WardName, Capacity, NurseInCharge)

ADMISSION(AdmissionID, PatientID, AdmissionDate, DischargeDate, Reason)

69. (a) Draw an entity-relationship diagram showing the three entities and the relationships between them. Show the cardinality of each relationship. Underline the primary key in each entity. [4]

(b) Write a SQL query to display the first name, last name, and ward name of all patients currently in the hospital. [3]

(c) The database is currently in third normal form (3NF). Explain TWO benefits of having the database in 3NF rather than as a single flat-file table. [4]

Benefit 1: _____

Benefit 2: _____

(d) Write a SQL query to find the number of patients in each ward.

Display the WardName and the count of patients. [1]

QUESTION 14 — Networks & Security (10 marks)

70. (a) A school has a star topology network. Explain what would happen to the network if:

(i) One student's computer cable is disconnected. [1]

(ii) The central switch fails. [2]

(b) Describe the difference between a router and a switch. [3]

(c) Explain how SSL/TLS encryption protects data sent between a web browser and a web server when a user enters their password. [4]

QUESTION 15 — Algorithms & Data Representation (8 marks)

71. (a) Describe the steps of the bubble sort algorithm. [3]

(b) A student records sound using a sampling rate of 44.1 kHz and 16-bit sampling resolution. Explain how each of these two values affects the quality of the recorded sound and the file size. [4]

Sampling rate: _____

Sampling resolution: _____

(c) State ONE advantage of using hexadecimal rather than binary to represent memory addresses. [1]

SECTION B TOTAL: 30 marks

SECTION C: Extended Answer (15 marks)

Answer ALL parts of the question.

QUESTION 16 — Synthesis Question (15 marks)

Victoria Park Academy Shenzhen is planning to build a new smart campus. The ICT manager has been asked to design the network infrastructure and security systems. The requirements are:

- 500 student computers across 20 classrooms
- 50 teacher computers in staff offices
- 20 wireless access points for student and teacher devices
- 10 network printers
- A central server room with file server, web server, and database server
- Internet access for all users
- All data must be protected from unauthorised access
- Students and teachers must have different levels of access to resources

72. (a) Draw a labelled network diagram showing the topology and all hardware devices you would recommend for this smart campus.

Justify your choice of topology. [5]

(b) Explain TWO security measures that should be implemented to protect the network and the data stored on the servers. [4]

Measure 1: _____

Measure 2: _____

(c) The database server stores student records including personal information. Explain how the principles of the Data Protection Act should be applied to this data. [4]

(d) The school wants to use cloud storage as a backup solution for critical data. Describe TWO advantages of using cloud storage compared to local backup solutions. [2]

Advantage 1: _____

Advantage 2: _____

SECTION C TOTAL: 15 marks

END OF MOCK EXAM

GRAND TOTAL: 75 marks

[TEACHER ONLY] MOCK EXAM — MARK SCHEME & GRADE BOUNDARIES

SECTION A: SHORT ANSWER (30 marks) — DETAILED MARK SCHEME

Question 1 — Binary to Hexadecimal (2 marks)

10011011

Split into nibbles: 1001 1011

1001 = 9, 1011 = 11 = B

Answer: 9B

- 1 mark for splitting correctly or showing working
- 1 mark for correct answer 9B

Question 2 — Image file size (3 marks)

Resolution: $800 \times 600 = 480,000$ pixels

Colour depth: 24 bits per pixel

Total bits: $480,000 \times 24 = 11,520,000$ bits

Convert to bytes: $11,520,000 \div 8 = 1,440,000$ bytes

Convert to MB: $1,440,000 \div (1024 \times 1024) = 1.37$ MB (or ~ 1.4 MB)

- 1 mark: Correct calculation of total pixels (800×600)
- 1 mark: Multiplied by colour depth (24 bits)
- 1 mark: Correct final answer in MB (accept 1.37 MB or ~ 1.4 MB)

(Accept method using bytes: $800 \times 600 \times 3 = 1,440,000$ bytes;

$1,440,000 \div 1,048,576 = 1.37$ MB)

Question 3 — Lossy vs lossless (2 marks)

- 1 mark: Lossy compression permanently removes some data, which would lose/corrupt text content making it unreadable
- 1 mark: Lossless compression reduces file size by finding patterns without removing any data, so all text is preserved exactly

Question 4 — DNS role (3 marks)

- 1 mark: DNS translates domain names into IP addresses
- 1 mark: When a user types a URL, the browser sends a request to a DNS server
- 1 mark: The DNS server returns the corresponding IP address so the browser can connect to the correct web server

Question 5 — Internet vs WWW (2 marks)

- 1 mark: The Internet is a global network of interconnected computers using standard protocols (TCP/IP)
- 1 mark: The World Wide Web is a collection of web pages and resources accessed via the Internet using HTTP/HTTPS

(Accept: Internet is the infrastructure; WWW is a service that runs on top of it)

Question 6 — Packet switching (4 marks)

- 1 mark: Data is broken into small packets
- 1 mark: Each packet can take a different route to the destination
- 1 mark: Packets are reassembled at the destination
- 1 mark: More reliable because if one route fails, packets can be re-routed / if one packet is lost or corrupted, only that packet needs to be resent, not the entire message

(Accept: comparison to circuit switching — packet switching shares bandwidth more efficiently)

Question 7 — CPU components (5 marks total)

(a) Three components (3 marks — 1 mark each):

- ALU (Arithmetic Logic Unit): performs arithmetic and logical calculations/operations
- CU (Control Unit): fetches, decodes, and executes instructions; controls the flow of data
- Registers: small, fast storage locations that hold data and instructions currently being processed

(b) Cache memory (2 marks):

- 1 mark: Cache is a small amount of very fast memory located inside/on the CPU
- 1 mark: It stores frequently used data/instructions so the CPU can access them faster than from RAM, reducing processing time

Question 8 — System vs application software (3 marks)

- 1 mark: System software manages the computer hardware and provides a platform for running application software / manages resources
- 1 mark: Application software is designed to help users perform specific tasks (e.g., word processing, web browsing)
- 1 mark: Example of system software (e.g., operating system, utility software, device drivers) AND example of application software (e.g., Microsoft Word, web browser, spreadsheet)

Question 9 — Firewall (2 marks)

- 1 mark: A firewall monitors incoming and outgoing network traffic
- 1 mark: It blocks/allows traffic based on security rules / acts as a barrier between a trusted network and untrusted networks (e.g., Internet)

Question 10 — Encryption (3 marks)

- 1 mark: Encryption is the process of converting plain text/data into an unreadable format (cipher text) using an algorithm and key
- 1 mark: Important because data sent over the Internet passes through many networks/routers where it could be intercepted
- 1 mark: Without encryption, intercepted data could be read by unauthorised people; encryption ensures only the intended recipient can decrypt and read it

Question 11 — Database keys (4 marks total)

(a) Primary keys (2 marks):

- STUDENT: PatientID (accept StudentID)
- CLASS: ClassID

(1 mark each)

(b) Link explanation (2 marks):

- 1 mark: The tables are linked by the ClassID field
- 1 mark: ClassID is the primary key in CLASS and a foreign key in STUDENT, creating a relationship between the two tables

Question 12 — Truth table (4 marks)

A	B	C	NOT B	A AND NOT B	B AND C	X
0	0	0	1	0	0	0
0	0	1	1	0	0	0
0	1	0	0	0	0	0
0	1	1	0	0	1	1
1	0	0	1	1	0	1
1	0	1	1	1	0	1
1	1	0	0	0	0	0
1	1	1	0	0	1	1

Mark allocation: 1 mark per column (NOT B, A AND NOT B, B AND C, X)

All 8 rows must be correct in each column for the mark.

[4 marks total]

SECTION B: STRUCTURED QUESTIONS (30 marks) — DETAILED MARK SCHEME

Question 13 — Databases & SQL (12 marks)

(a) ERD — 4 marks:

- 1 mark: Three entities drawn and correctly named (PATIENT, WARD,

ADMISSION)

- 1 mark: All attributes listed with primary keys underlined
(PatientID, WardID, AdmissionID)
- 1 mark: Correct relationships drawn with appropriate direction
(PATIENT linked to WARD; PATIENT linked to ADMISSION)
- 1 mark: Correct cardinalities shown:
WARD 1:M PATIENT (one ward has many patients)
PATIENT 1:M ADMISSION (one patient can have many admissions)

[4 marks total]

(b) SQL query — 3 marks:

```
SELECT PATIENT.FirstName, PATIENT.LastName, WARD.WardName
FROM PATIENT
INNER JOIN WARD ON PATIENT.WardID = WARD.WardID; -- 3 marks
```

Mark allocation:

- 1 mark: Correct SELECT with three columns (accept * or any variation that includes these fields)
- 1 mark: Correct FROM and INNER JOIN syntax
- 1 mark: Correct ON condition linking PATIENT.WardID = WARD.WardID
(Accept WHERE PATIENT.WardID = WARD.WardID in place of ON clause)

[3 marks total]

(c) 3NF benefits — 4 marks (2 marks each):

Benefit 1:

- 1 mark: Reduced data redundancy / no repeated data
- 1 mark: Explanation — data is stored in only one place, so less storage space is needed / updates only need to be made in one place

OR

Benefit 2:

- 1 mark: Improved data integrity / consistency
- 1 mark: Explanation — because data is not duplicated, there is no risk of inconsistent values / anomaly prevention (insertion, update, deletion anomalies)

OR

Benefit 3:

- 1 mark: Easier to maintain / modify the database structure
- 1 mark: Explanation — changes to one table do not affect others unnecessarily / tables can be modified independently

[4 marks total]

(d) SQL count query — 1 mark:

```
SELECT WARD.WardName, COUNT(*) AS PatientCount
FROM PATIENT
INNER JOIN WARD ON PATIENT.WardID = WARD.WardID
GROUP BY WARD.WardName;
```

(Accept any functionally correct query; GROUP BY is essential)

[1 mark total]

Question 14 — Networks & Security (10 marks)

(a)(i) One student's cable disconnected — 1 mark:

Only that student's computer loses connection. The rest of the network continues to function normally because each device has its own dedicated connection to the central switch.

(a)(ii) Central switch fails — 2 marks:

- 1 mark: The entire network fails / all devices lose connection
- 1 mark: Because all data passes through the central switch; there is a single point of failure

(b) Router vs switch — 3 marks:

- 1 mark: A switch connects devices within the same local network and forwards data to the correct destination device using MAC addresses
- 1 mark: A router connects different networks together (e.g., LAN to Internet) and routes data between them using IP addresses
- 1 mark: A router can perform Network Address Translation (NAT) and assign IP addresses; a switch operates at the data link layer while a router operates at the network layer

(c) SSL/TLS encryption — 4 marks:

- 1 mark: SSL/TLS encrypts data transmitted between the browser and the web server
- 1 mark: When a password is entered, it is encrypted by the browser before being sent
- 1 mark: Even if the data is intercepted, it cannot be read without the decryption key
- 1 mark: The website's identity is verified using a digital certificate, preventing man-in-the-middle attacks

(Accept: HTTPS uses port 443; creates a secure "tunnel")

Question 15 — Algorithms & Data Representation (8 marks)

(a) Bubble sort steps — 3 marks:

- 1 mark: Compare adjacent items in the list
- 1 mark: If they are in the wrong order, swap them
- 1 mark: Repeat this process for multiple passes until no more

swaps are needed / the list is sorted

(Accept: Continue until a complete pass with no swaps)

(b) Sound recording — 4 marks:

Sampling rate (44.1 kHz) — 2 marks:

- 1 mark: The sampling rate determines how many samples of the sound are taken per second (44,100 samples/sec)
- 1 mark: Higher sampling rate means better quality (captures more detail of the original sound) but larger file size

Sampling resolution (16-bit) — 2 marks:

- 1 mark: The sampling resolution determines the number of bits used to store each sample / the number of discrete levels (65,536)
- 1 mark: Higher resolution means better quality (more accurate representation of amplitude) but larger file size

(c) Hexadecimal advantage — 1 mark:

- Hexadecimal is more compact/shorter than binary (one hex digit = four binary digits)
- Easier for humans to read and write
- Less prone to transcription errors

(Any ONE valid advantage — 1 mark)

SECTION C: EXTENDED ANSWER (15 marks) — DETAILED MARK SCHEME

Question 16 — Smart Campus Synthesis (15 marks)

(a) Network diagram and topology justification — 5 marks:

Diagram — 3 marks:

- 1 mark: Star topology shown (all devices connect to central points)
- 1 mark: Key hardware shown — at least one central switch, router, and server room / servers
- 1 mark: All required elements present (student computers, teacher computers, access points, printers, servers, internet connection)

Topology justification — 2 marks:

- 1 mark: Star topology chosen
- 1 mark: Justified by any TWO valid reasons:
 - * Easy to add/remove devices (scalable for 500+ devices)
 - * Fault isolation — one cable failure doesn't affect others
 - * Centralised management — easy to monitor and control

* High performance — dedicated bandwidth per connection

* Easy to troubleshoot

(b) Security measures — 4 marks (2 marks each):

Measure 1:

- 1 mark: Firewall installed between network and Internet
- 1 mark: Monitors and filters incoming/outgoing traffic based on security rules; blocks unauthorised access / malware

OR

- 1 mark: User authentication system (username/password or biometric)
- 1 mark: Ensures only authorised users can access resources; different access levels for students and staff

OR

- 1 mark: Encryption of sensitive data (SSL/TLS for data in transit, disk encryption for data at rest)
- 1 mark: Protects data even if intercepted or if physical storage is stolen

OR

- 1 mark: Regular backups stored offsite / in cloud
- 1 mark: Ensures data can be recovered in case of ransomware, hardware failure, or disaster

(c) Data Protection Act principles — 4 marks:

- 1 mark: Data must be collected fairly and lawfully / users must be informed about what data is collected and why
- 1 mark: Data must only be used for the purpose for which it was collected (e.g., student records only for educational purposes)
- 1 mark: Data must be kept secure — access controls, encryption, passwords to prevent unauthorised access
- 1 mark: Data must be accurate and kept up to date / not kept longer than necessary / individuals have the right to access their own data

(Accept any valid Data Protection Act principles — these are generic principles applicable to most data protection legislation)

(d) Cloud storage advantages — 2 marks (1 mark each):

Advantage 1:

- 1 mark: Accessible from anywhere with an Internet connection
- 1 mark: Explanation — staff can access files remotely; disaster recovery if local site is unavailable

OR

Advantage 2:

- 1 mark: Automatic backups / data redundancy
- 1 mark: Cloud providers replicate data across multiple servers/ locations, reducing risk of data loss

OR

Advantage 3:

- 1 mark: Scalable storage / pay for what you use
- 1 mark: School can increase/decrease storage as needed without buying physical hardware

OR

Advantage 4:

- 1 mark: Reduced need for on-site maintenance / IT staff
- 1 mark: Cloud provider manages hardware, updates, and security

MOCK EXAM GRADE BOUNDARIES & TOTALS

Total marks available: 75

CIE-style grade boundaries:

A*: 67–75 marks (89% and above)

A: 60–66 marks (80–88%)

B: 53–59 marks (71–79%)

C: 46–52 marks (61–70%)

D: 39–45 marks (52–60%)

E: 32–38 marks (43–51%)

U: 0–31 marks (below 43%)

Adjustments: Teachers may adjust boundaries by ± 3 marks based on cohort performance, question difficulty analysis, and comparison to CIE past paper boundaries.

Record sheet columns for teacher use:

[Student Name] | [Sec A /30] | [Sec B /30] | [Sec C /15] | [Total /75] | [Grade]

STUDENT REPORT TEMPLATE

VICTORIA PARK ACADEMY SHENZHEN

CIE IGCSE Computer Science (0478) — Year 10 Semester 1

STUDENT PROGRESS REPORT

Student Name: _____ Class: _____

Report Date: _____ Teacher: _____



1. ATTAINMENT SUMMARY

||

A horizontal timeline bar with tick marks for the years 2014, 2015, 2016, and 2017. A bracket below the bar spans from 2014 to 2015, indicating the duration of the study.

Assessment → Max Marks → Raw Score → Percentage

Quiz 1: Databases & SQL → 25

(Week 3)

Quiz 2: Boolean Logic → 20

(Week 5)

Quiz 3: Sorting & Validation → 20

(Week 7)

Quiz 4: Network Hardware → 20

(Week 8)

Mock Exam: Paper 1 → 75

(Week 17)

RUNNING TOTAL (all assessed) → 160

Overall grade for semester: _____

Grade descriptor:

A* = 90%+ A = 80-89% B = 70-79% C = 60-69%

D = 50-59% E = 40-49% U = below 40%

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2. TOPIC BREAKDOWN

Rate each topic: E = Emerging D = Developing S = Secure

NEW CONTENT (Y10S1)

Topic → Rating → Notes

1. Databases — concepts → E / D / S

(tables, records, fields, keys)

2. Databases — relationships → E / D / S

(1:1, 1:M, M:N, ERDs)

3. Databases — normalisation → E / D / S

(1NF, 2NF, 3NF)

4. SQL — SELECT, WHERE, ORDER BY → E / D / S

5. SQL — JOIN, INSERT, UPDATE → E / D / S

6. Boolean Logic — gates → E / D / S
(AND, OR, NOT, NAND, NOR, XOR)
7. Boolean Logic — truth tables → E / D / S
8. Boolean Logic — circuits → E / D / S
9. Bubble sort algorithm → E / D / S
10. Bubble sort — trace tables → E / D / S
11. Validation checks → E / D / S
12. Network hardware (NIC, MAC, IP) → E / D / S
13. Network topologies → E / D / S
14. Client-server vs P2P → E / D / S

REVISION TOPICS (Y9)

Topic	Rating	Notes
15. Data Representation	→ E / D / S	(binary, hex, ASCII, images)
16. Communication & Internet	→ E / D / S	
17. Hardware (CPU, memory, storage)	→ E / D / S	
18. Software (OS, applications)	→ E / D / S	
19. Security & Ethics	→ E / D / S	
20. Algorithm Design	→ E / D / S	
21. Programming constructs	→ E / D / S	

Topic rating definitions:

- E (Emerging) = Can recognise concepts with support; cannot yet apply independently. Priority area for Y10S2.
- D (Developing) = Can apply knowledge to familiar problems; some errors under exam conditions. Continue to practise.
- S (Secure) = Can apply knowledge confidently to unfamiliar problems; exam-ready. Maintain through spaced practice.

3. STRENGTHS		
4. AREAS FOR DEVELOPMENT		

5. TARGETS FOR YEAR 10 SEMESTER 2	

Target 1 (highest priority):

Target 2:

Target 3:

Teacher comments:

Student signature: _____ Date: _____

Parent/Guardian signature: _____ Date: _____

SUGGESTED COMMENT BANK (for teacher use — copy and adapt as needed)

STRENGTHS — New Content:

- Excellent understanding of relational database concepts
- Accurate and confident SQL query writing
- Strong grasp of all six logic gates and their truth tables
- Can convert between logic expressions and circuits accurately
- Thorough understanding of bubble sort — can trace correctly every time
- Good application of validation checks in Python code
- Clear understanding of network hardware and their roles
- Can design appropriate network diagrams with correct topology choices

STRENGTHS — Revision Content:

- Binary/hexadecimal conversions are secure and quick
- Sound understanding of how the Internet and WWW work
- Can explain CPU components and their functions clearly
- Good recall of security measures and when to apply them

AREAS FOR DEVELOPMENT:

- Needs more practice writing SQL queries with JOIN operations
- Truth tables with three inputs need more careful checking
- Bubble sort trace tables — work through more slowly, check each swap
- Validation code — practise writing from memory without templates
- Network topology comparisons — memorise advantages/disadvantages
- Exam timing — practise working under timed conditions
- Extended answer questions — need to develop more detailed explanations

TARGETS:

- Complete at least 3 past paper questions per week over the holiday
- Create personal flashcards for all SQL keywords and Boolean gates
- Practise bubble sort traces with arrays of different sizes
- Review Y9 topics for 15 minutes every day using the revision folder

- Attempt the February/March 2023 Paper 1 under timed conditions

BILINGUAL GLOSSARY POSTER — YEAR 10 SEMESTER 1

UNIT 1 — DATABASES & SQL (数据库与 SQL)

English → Chinese → Pinyin

database → 数据库 → shujukù

table → 表 → biǎo

record → 记录 → jìlù

field → 字段 → zìduàn

primary key → 主键 → zhǔjiàn

foreign key → 外键 → wàijiàn

flat-file database → 平面文件数据库 → píngmiàn wénjiàn shujukù

relational database → 关系型数据库 → guānxi xíng shujukù

entity → 实体 → shí tǐ

relationship → 关系 → guānxi

entity-relationship → 实体关系图 → shí tǐ guānxi tú

diagram (ERD)

cardinality → 基数 → jīshù

one-to-one (1:1) → 一对一 → yī duì yī

one-to-many (1:M) → 一对多 → yī duì duō

many-to-many (M:N) → 多对多 → duō duì duō

normalisation → 规范化 → guīfānhuà

first normal form (1NF) → 第一范式 → dì yī fānshì

second normal form (2NF) → 第二范式 → dì èr fānshì

third normal form (3NF) → 第三范式 → dì sān fānshì

redundancy → 冗余 → róngyú

data integrity → 数据完整性 → shùjù wánzhèng xìng

SQL → 结构化查询语言 → jiégòuhuà chāxún yǔyán

SELECT → 选择/查询 → xuǎnzé / chāxún

FROM → 从... → cóng

WHERE → 条件/哪里 → tiáojiàn / nǎlǐ

ORDER BY → 排序 → páixù

ASC / DESC → 升序/降序 → shēngxù / jiàngxù

wildcard → 通配符 → tōngpèifú

INSERT → 插入 → charu
UPDATE → 更新 → gengxin
DELETE → 删除 → shanchu
JOIN → 连接 → lianjie
INNER JOIN → 内连接 → nei lianjie
query → 查询 → chaxun

UNIT 2 — BOOLEAN LOGIC (布尔逻辑)

English → Chinese → Pinyin
Boolean → 布尔 → buer
Boolean algebra → 布尔代数 → buer daishu
logic gate → 逻辑门 → luoji men
AND gate → 与门 → yu men
OR gate → 或门 → huo men
NOT gate → 非门 → fei men
NAND gate → 与非门 → yu fei men
NOR gate → 或非门 → huo fei men
XOR gate → 异或门 → yi huo men
truth table → 真值表 → zhenzhi biao
logic expression → 逻辑表达式 → luoji biaodashi
logic circuit → 逻辑电路 → luoji dianlu
input → 输入 → shuru
output → 输出 → shuchu
Boolean variable → 布尔变量 → buer bianliang
truth value → 真值 → zhenzhi
true → 真/正确 → zhen / zhengque
false → 假/错误 → jia / cuowu
half-adder → 半加器 → banjia qi
full-adder → 全加器 → quanjia qi
De Morgan's laws → 德摩根定律 → De Mogen dinglu

UNIT 3 — SORTING & VALIDATION (排序与验证)

English → Chinese → Pinyin

bubble sort → 冒泡排序 → maopao paixu

sorting algorithm → 排序算法 → paixu suanfa

algorithm → 算法 → suanfa

pass → 趟/轮 → tang / lun

swap → 交换 → jiaohuan

compare → 比较 → bijiao

sorted → 已排序 → yi paixu

unsorted → 未排序 → wei paixu

efficiency → 效率 → xiaolu

Big O notation → 大 O 表示法 → da O biaooshifa

linear search → 顺序搜索 → shunxu sousuo

binary search → 二分搜索 → erfen sousuo

validation → 验证 → yanzheng

verification → 校验 → jiaoyan

range check → 范围检查 → fanwei jiancha

type check → 类型检查 → leixing jiancha

length check → 长度检查 → changdu jiancha

presence check → 存在检查 → cunzai jiancha

format check → 格式检查 → geshi jiancha

check digit → 校验位 → jiaoyan wei

parity check → 奇偶校验 → ji'ou jiaoyan

UNIT 4 — NETWORK HARDWARE (网络硬件)

English → Chinese → Pinyin

Network Interface Card → 网络接口卡 → wangluo jiekou ka
(NIC)

MAC address → MAC 地址/物理地址 → MAC dizhi / wuli dizhi

IP address → IP 地址 → IP dizhi

IPv4 → IPv4 → IPv4

router → 路由器 → luyou qi

switch → 交换机 → jiaohuan ji

access point → 接入点 → jieru dian

network topology → 网络拓扑 → wangluo tuopu

star topology → 星型拓扑 → xing xing tuopu
bus topology → 总线拓扑 → zongxian tuopu
mesh topology → 网状拓扑 → wangzhuang tuopu
hybrid topology → 混合拓扑 → hunhe tuopu
client-server → 客户机-服务器 → kehuji-fuwuqi
peer-to-peer (P2P) → 点对点 → dian dui dian
packet → 数据包 → shuju bao
protocol → 协议 → xieyi
bandwidth → 带宽 → daikuan
latency → 延迟 → yanchi
Ethernet → 以太网 → yitai wang
Wi-Fi → 无线局域网 → wuxian juyu wang
firewall → 防火墙 → fanghuo qiang
encryption → 加密 → jiami
decryption → 解密 → jiemi

YEAR 9 REVISION TERMS (Y9 复习词汇)

English → Chinese → Pinyin
binary → 二进制 → erjinzhi
hexadecimal → 十六进制 → shiliu jinzhi
denary / decimal → 十进制 → shijinzhi
ASCII → ASCII 码 → ASCII ma
pixel → 像素 → xiangsu
resolution → 分辨率 → fenbianlu
sampling rate → 采样率 → caiyanglu
sampling resolution → 采样分辨率 → caiyang fenbianlu
bit depth → 位深度 → wei shendu
compression → 压缩 → yasuo
lossy compression → 有损压缩 → yousun yasuo
lossless compression → 无损压缩 → wusun yasuo
Internet → 互联网 → hulianwang
World Wide Web (WWW) → 万维网 → wanweiwang
URL → 统一资源定位符 → tongyi ziyuan dingwei fu
HTTP / HTTPS → 超文本传输协议 → chaowenben chuanshu xieyi
DNS → 域名系统 → yuming xitong

packet switching → 分组交换 → fenzu jiaohuan
circuit switching → 电路交换 → dianlu jiaohuan
CPU → 中央处理器 → zhongyang chuli qi
ALU → 算术逻辑单元 → suanshu luoji danyuan
CU → 控制单元 → kongzhi danyuan
register → 寄存器 → jicun qi
cache → 缓存 → huancun
RAM → 随机存取存储器 → suiji cunqu cunchu qi
ROM → 只读存储器 → zhidu cunchu qi
virtual memory → 虚拟内存 → xuni neicun
operating system → 操作系统 → caozuo xitong
malware → 恶意软件 → eyi ruanjian
phishing → 网络钓鱼 → wanglu diaoyu
hacking → 黑客攻击 → heike gongji
backup → 备份 → beifen
encryption → 加密 → jiami
private key → 私钥 → siyao
public key → 公钥 → gongyao
variable → 变量 → bianliang
constant → 常量 → changliang
assignment → 赋值 → fuzhi
selection → 选择 → xuanze
iteration → 迭代/循环 → diedai / xunhuan
array → 数组 → shuzu
subroutine → 子程序 → zicheng xu
procedure → 过程 → guocheng
function → 函数 → hanshu
parameter → 参数 → canshu
pseudocode → 伪代码 → weidaima
trace table → 追踪表 → zhuizong biao

EXAM COMMAND WORDS (考试指令词)

English → Chinese → Pinyin

State → 陈述 → chenshu

Identify → 识别 → shibie

Describe → 描述 → miaoshu

Explain → 解释 → jieshi

Compare → 比较 → bijiao

Evaluate → 评估 → pinggu

Justify → 论证 → lunzheng

Calculate → 计算 → jisuan

Draw / Sketch → 绘制 → huizhi

Write → 写出 → xiechu

Give / State one → 给出一个... → geichu yige

--- TOTAL TERMS IN POSTER: 105 terms across 5 units + exam command words ---

PRINTING INSTRUCTIONS FOR CLASSROOM DISPLAY

This glossary poster is formatted for A3 printing (portrait or landscape).

Recommended display method:

73. Print each unit on a separate A3 sheet

74. Laminate for durability

75. Display in sequence on classroom wall

76. Add new units as they are taught (cumulative display)

77. Keep Year 9 revision terms visible at all times

Alternative: Print at A4 and combine into student booklet (5 pages).

END OF ASSESSMENT MATERIALS & REPORTING PACK — Year 10 Semester 1

Victoria Park Academy Shenzhen — CIE IGCSE Computer Science (0478)